Indian Institute of Technology, Gandhinagar

EE226 Semiconductor Devices

Semester 2, 2025-2026

Tutorial – 5

Q1. **Read Section 4.1 (Constancy of Fermi Level) from "Fundamentals of Semiconductor Devices" by B. Anderson & R. Anderson**

Describe why the Fermi level should remain constant (same level) in thermal equilibrium when two different materials are brought together to form a junction?

Q2. Describe the steps for drawing a band diagram when two materials are brought together. Which energy levels accommodate the difference in Fermi levels of these two materials?

Q3. What happens to the Fermi level when an applied voltage $V_A \neq 0$. Where will the applied voltage be observed in the band diagram?

Q4. For uniform and non-uniform semiconductors with no voltage applied:

a. Find the drift current, diffusion current and total current.

b. Describe electron and hole flow directions for drift, diffusion

c. Describe the current flow direction for drift current, diffusion current, and total current.

Q5. Refer class notes Draw the thermal equilibrium band diagram of the following junctions by marking the work function, barrier heights, and $EF - Ei$ difference using the given information: (N_a=10^{17}cm-3, N_d=10^{18}cm-3, kT/q = 26mV, metal work function = 5eV, electron affinity of semiconductor = 4eV, bandgap of semiconductor = 1.1eV, oxide electron affinity = 1eV, oxide band gap = 9eV.

a. P-type semiconductor (P)

b. N-type semiconductor (N)

c. PN junction

d. P/intrinsic/N (PIN)

e. NPN junction

f. MOS (Metal/Oxide/Semiconductor) (with N and P both)

g. MS (Metal/Semiconductor) junction (with N and P both)

h. M1/Oxide/M2 i. with same function

ii. With $\Delta\phi$ = $1eV$ iii. With $\Delta\phi$ = $-1eV$

i. Draw an approximate band diagram for an N-type semiconductor (Eg_1)/N-type semiconductor (Eg_2) with.

i. Eg_1>Eg_2

ii. Eg_2>Eg_1

Note: There can be multiple answers in terms of Ec & Ev level difference. Choose any one case of your choice and draw.

Q6. Describe the shape (abrupt, continuous) of Ec, Ev, and Ei exactly at the interface of two materials when brought together (form junction)? Which case will have abrupt changes and which will have continuous change?

Q7. With a flowchart, write the relationship between band diagram, Potential, Electric Field, Charge density and carrier concentration. Draw the approximate Charge density, Electric Field, Potential profiles and Band Diagram for the following structures. Take suitable assumptions while drawing. (No need of exact derivations except PN Junction)

> a. PN Junction Diode Read complete section 5.2 (Equilibrium Conditions) of Streetman book
>
> b. MS Junction Diode
>
> c. NPN
>
> d. MOS

Q8. For a particular junction, the acceptor concentration is N_a=10^{16} cm-3 and the donor concentration is N_d =10^{15}cm-3

a. Find the junction's built-in potential (V_{bi}). Assume ni =10^{10}cm-3

b. Find depletion width (W) and its extent in each of the p and n regions i.e. x_p and x_n.

c. When the junction is reverse bias with reverse applied voltage 5 V. Find x_p and x_n and W at this RB voltage of 5V.

d. Calculate the magnitude of the charge stored on either side of the junction. Assume the junction area is 400 μm^2.

Q9.Few questions from the book **"Fundamentals of Semiconductor Devices" by B. Anderson & R. Anderson**

5.1 A silicon pn junction is formed between n-type silicon doped with $N_D = 5.0 \times 10^{15}$ cm^{-3} and p-type silicon doped with $N_A = 2.0 \times 10^{17}$ cm^{-3}.

a. Sketch the energy band diagram. Label both axes and all important energy levels.

b. Find n_{n0}, p_{n0}, n_{p0}, and p_{p0}. Sketch the carrier concentrations.

c. What is the built-in voltage?

5.5 A step pn junction diode is made in silicon with the n side having $N'_D = 2 \times 10^{16}$ cm^{-3} and the p side having a net doping of $N'_A = 5 \times 10^{15}$ cm^{-3}.

a. Draw, to scale, the energy band diagram of the junction at equilibrium.

b. Find the built-in voltage, and compare with the value measured off your drawing in part (a).

c. Find the junction width.

d. Find the widths of the n side of the depletion region and the p side of the depletion region, and the voltage dropped across each side of the transition region.

e. Plot the electric field. What is its maximum value?

f. Plot the voltage distribution.

g. Plot the potential energy for electrons (E_C).

h. Draw the energy band diagram for $V_a = 0.5$ V.

i. Draw the energy band diagram for $V_a = -5$ V.

5.6 Consider the equilibrium energy band diagram for the pn junction diode shown in Figure P5.1a.

a. Indicate the region(s) where there exists an electric field.

b. What is the value of the built-in voltage?

c. Sketch the electron and hole concentrations. Indicate the directions of the drift and diffusion components of the electron and hole fluxes and currents.

d. The same device is shown in Figure P5.1b, but now a voltage is applied across it. If the total junction voltage $V_j = V_{bi} - V_a$, what is the value of the applied voltage?

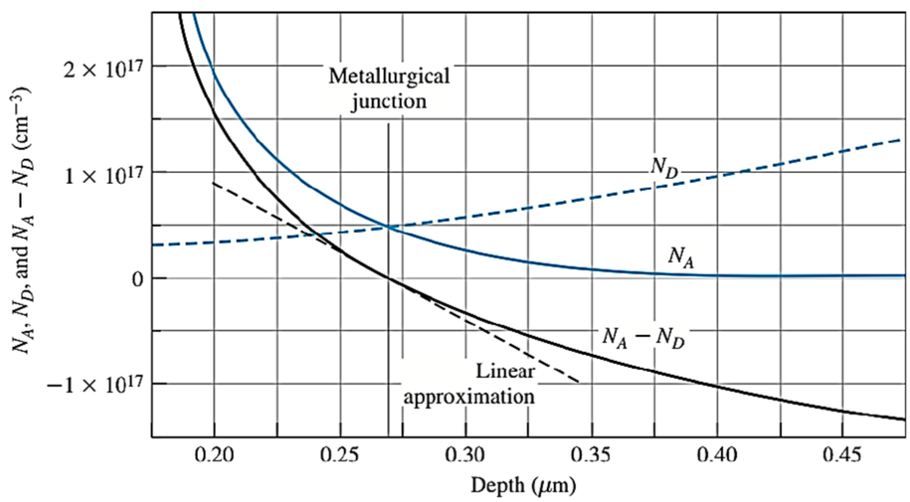
Q10.A silicon step junction maintained at room temperature is doped such that Ef – Ev =2kT on the p side and Ef = Ec – Eg/4 on the n-side.

(a) Draw the equilibrium energy band diagram for this junction.

(b) Obtain the built-in voltage giving both a symbolic and a numerical result.

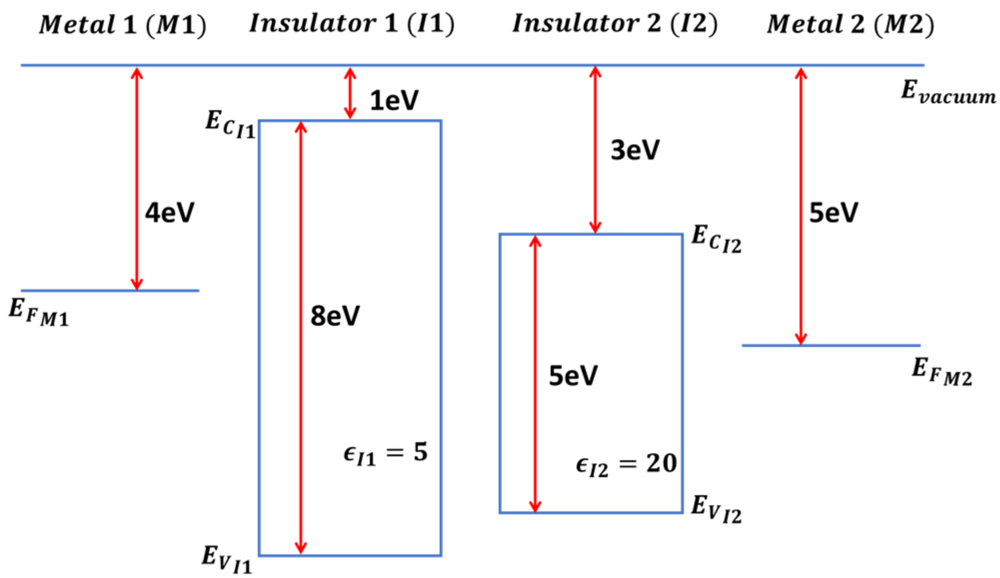
Q11**.** During the fabrication of PN junction, it was realized that doping for P-type (N_A) and N-type (N_D) semiconductor was not uniform. The exact doping profiles during fabrication of PN junction are shown in figure below. The metallurgical junction is the boundary of PN junction at x_0. It can be observed that, the $(N_A - N_D)$ is not uniform.

For simplicity, assume that $(N_A - N_D)$ follows a linear change approximation near the junction defined by $(N_A - N_D) = -a(x - x_0)$ as shown in figure, where $'a'$ is the magnitude of slope. You may assume $x_0 = 0$ for simplicity and use depletion approximation.



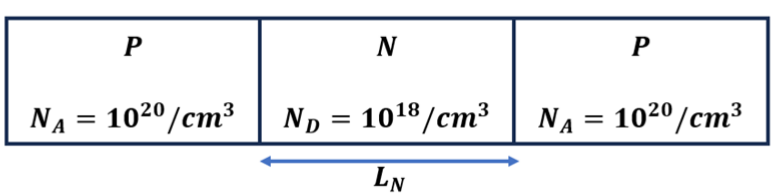
a. Derive the expression for depletion width (W). Describe the W dependency on $'a'$
b. Derive the expression for built-in voltage. Can built in voltage be solved directly?
c. Suppose junction voltage $V_j = V_{bi} - V_a = 1V$, calculate the depletion width? Is linear change approximation valid?
d. Suppose junction voltage $V_j = V_{bi} - V_a = 10V$, calculate the depletion width? Is linear change approximation valid?
e. Draw charge density, E- field, potential and band-diagram. Annotate different regions/parameters clearly.

Q12. Four different materials and parameters are shown below before junction formation. Assume the length of all the materials is 50nm each. When they are connected together M1/I1/I2/M2, under thermal equilibrium:



a. Draw the energy band diagram. Specify exact barrier heights, band bending in different regions.
b. Potential profile. Specify exact potential drop in different regions.
c. Electric field profile. Specify exact Electric Field value in different regions.
d. Charge density profile.

Q13. For a PNP structure shown below, assume the N-region length (L_N) is sufficiently large so that the depletion regions do not merge from two junctions. Assume Depletion approximation.



a. Calculate the potential barrier value
b. Calculate depletion width (in nm) in P-region and N-region. Which region has larger portion of total depletion width?
c. Draw Band diagram, potential, E-field and charge profile.

Now, assume $L_N = 30nm$. You will observe that the depletion regions from both sides merge inside N-region.

d. Derive the equation for the new potential barrier and show its dependence on N_D, L_N
e. Calculate the new potential barrier value.
f. Draw the Band diagram, potential, E-field and charge profile when $L_N = 30nm$. Mark peak barrier value, peak electric field value.
g. With $L_N = 30nm$, to achieve the peak potential barrier of 0.5V, what should be N_D?
h. With $N_D = 10^{18}/cm^3$, to achieve the peak potential barrier of 0.5V, what should be the L_N in nm?

Following questions are part of the tutorial and the concept for them will be covered in next lecture.

Q14 For a PN junction diode:

a. How much change in voltage required to change the diode ideal current by one decade (10 times) i.e. $\frac{\mathrm{dV}}{\mathrm{d}(\log_{10}\mathrm{I})}$ (mV/decade) for a PN junction diode.

Assume $V_{bi} = 0.7\mathrm{V}$ and the current at $V_a = 0.7V$ is fixed at 1mA for a diode.

b. How much would be the leakage current at $V_a = 0.1V$ at room temperature T = 27 C?

c. If the diode is used in high temperature environment where $\mathrm{T} = 100\ \mathrm{C}$, how much would be the leakage current?

Q15. Many RF circuits employ voltage-controlled oscillators (VCOs) to select a specific frequency, with the frequency given by $f = \frac{1}{2\pi\sqrt{LC}}$. These oscillators typically use tank (LC) circuits, where the inductance (L) remains constant while the capacitance (C) can be varied by adjusting the applied voltage. PN junction diodes are commonly used as variable capacitors in these circuits in reverse bias. Consider, a PN junction where the p-side doping concentration $N_A = 10^{14}\ cm^{-3}$ and the n-side doping concentration $N_D = 10^{16}\ cm^{-3}$. Assume inductance $L = 50\ nH$.
Assume area of diode $100 \times 100\ \mu m^2$

a. Calculate the frequency of oscillator when applied voltage (V_a) = 0?
b. Calculate the reverse bias voltage needed to increase the frequency to 100 GHz? Is this a practical voltage?
c. If the maximum electric field allowable in the junction is $E_{max} \approx 10^5\ V/cm$ (breakdown limit), what is the maximum depletion width possible? What is the voltage corresponding to this maximum depletion width & find the maximum achievable frequency using this circuit?

You would have observed that there will be a limit on applied Voltage and E-field which would limit the maximum achievable frequency for the above given values of doping, inductance and area.

Suppose Area of the diode can be changed, now, if we need 100GHz frequency at $V_a = -10V$, what should be the area of the diode in μm^2?

Q16. A PIN diode i.e. P-type Si Semiconductor/Intrinsic/N-type Si Semiconductor is widely used in photodetectors, high voltage power electronics applications

a. Derive the current expression for PIN diode for:

   <u>Case1</u>: When intrinsic region is small compared to diffusion lengths $(L_i < (L_p, L_n))$

<u>Case2</u>: When intrinsic region is large compared to diffusion lengths $(L_i > (L_p, L_n))$

b. Draw charge density, E- field, potential and band-diagram for Case 2
   Annotate different regions/parameters clearly.